

PAPER_525 — Session 141 Hub: Universal Spectrum Spectral Divisions, DPM Frequency Drive, Quantum Egg Simulation, and Proplyd Bidirectional Framework

Unified Quantum Field Framework (UQFF)

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grok_share_3b6f26809.txt **CP4 Class:** Session141ProplydDPMSpectraHubCalculator (#120)

§1 — Session Overview

Source document: [grok_share_3b6f26809.txt](#) **Origin:** BigBangHypergraphTheory_12Dec2025.docx
— continued **Position in pipeline:** Continuation of Session 140 (DPM Shell Radiance, Negative Time, Force
Unification).

Papers generated: PAPER_521–525 (5 papers) **CP4 classes introduced:** #116–#120 (5 classes including this
hub)

§2 — Physics Advances Over Session 140

#	Advance	Session 140		Session 141 Upgrade
1	Spectral framework	Binary (attract/repel)	Ug1_dual	US overlay: 1/3 stable / 2/3 destructive
2	DPM role	Static parameters	DPM_dict	Quantum frequency driver DPM_drive
3	Big Bang echo	Not modeled		\$ReRing_{BB}\$: persistent Freq_max echo
4	Vacuum proof	Qualitative		$V_{\text{vacuum}} = \int Freq_{\text{open}} \cdot (E_{\text{egg}} - \text{Collapse}) \cdot P$
5	\$Ug1\$ field	$Ug1_{\text{dual}}$ (two modes)		$Ug1_{\text{spectra}}$ (simultaneous ranges)
6	UQFF tensor	Diagonal only	components	Spectral division tensor with off-diagonal couplings
7	Quantum egg	Qualitative description		Full numerical t_{neg} simulation
8	Plasma orb	Not defined		Probabilistic emergence threshold
9	Proplyd explanation	Not connected		Bidirectional DPM ↔ Proplyd framework
10	Centrip/Centrif forces	DPM-emergent (Session 140)		+ = US spectral weights 1/3 attract / 2/3 repel

§3 — New CP4 Classes

PAPER_521 — Universal Spectrum Spectral Divisions

CP4 #116 — UniversalSpectrumSpectralDivisionsCalculator

$$US^{(\text{range})} = \int Freq_{\text{drive}}, dt_{\text{neg}} \cdot \left(\frac{1}{3} A + O + \frac{2}{3} D \right) + ReRing_{BB}$$

$$V_{\text{vacuum}} = Freq_{\text{open}} \cdot (E_{\text{egg}} - \text{Collapse}) \cdot Prob_{\text{order}}$$

PAPER_522 — DPM as Quantum Frequency Driver

CP4 #117 — DPMFrequencyDriveReRingVacuumGradCalculator

$$DPM_{\text{drive}} = \frac{\kappa(DPM_n - DPM_s)}{r^{26}} \cdot US_{\text{overlay}}$$

$$\cdot \frac{\partial^{26} \text{Grind}_{\text{opp}}}{\partial t_{\text{adj}}^{26}} + \text{ReRing}_{\text{BB}}$$

$$Ug1_{\text{spectra}} = \frac{\partial^{26}(DPM_{\text{drive}})}{\partial r^{26}} \cdot \left(\frac{1}{3} A - \frac{2}{3} R \right) \cdot \text{ReRing}_{\text{BB}}$$

PAPER_523 — Quantum Egg Frequency Numerical Simulation**CP4 #118 — QuantumEggFrequencyNumericalSimCalculator**

$$US_{\text{egg}} = \int_{t_{\text{neg,min}}}^{0} f(t_{\text{neg}}) dt_{\text{neg}} \quad (\text{trapezoidal, 200 pts})$$

Validated: ALMA 225–345 GHz, exoALMA 230 GHz, VLA 92 GHz, JWST 0.97–5.27 μm, Hubble/MUSE 250–500 AU.

PAPER_524 — Plasma Orb Emergence Threshold**CP4 #119 — PlasmaOrbEmergenceThresholdCalculator**

$$US_{\text{orb}} > \mu + \sigma \cdot \text{Prob}_{\text{order}} \quad \rightarrow \quad \text{plasma orb emerges}$$

$$\text{Buoy}_{\text{grad}} = \frac{\rho_{\text{UA}} \cdot V_d \cdot (F_{\text{inert}} + R_h)}{1 + \Delta_{\text{dil}}}$$

Orion emergence fraction: **18.32%** | Mean size: **375.87 AU**

§4 — Proplyd ↔ DPM Bidirectional Framework

Proplyds explain DPM: Proplyds (proto-stellar disks undergoing photoevaporation) are physical realizations of the split monopole structure — the ionized outer disk corresponds to DPM_s (south/CCW) and the dense inner core to DPM_n (north/CW). The bipolar outflow is the observable signature of $\omega_{\text{CW}} - \omega_{\text{CCW}}$ differential rotation.

DPM explains Proplyds: The magnetic braking catastrophe in classical star formation (angular momentum of infalling gas is too large) is resolved by the DPM_{drive} frequency gradient:

$$\dot{J}_{\text{DPM}} = -\frac{\partial(DPM_{\text{drive}} \cdot r^{26})}{\partial r} \cdot L$$

DPM extracts angular momentum via the 26th-order spatial gradient, naturally reducing J to observed proplyd rotation rates without requiring a singular disk/wind coupling.

§5 — Three UQFF Number Systems — New Contexts (PAPER_429)

System	PAPER_429 Definition	Session 141 New Context
Vacuum Density Series	$\$Li_{\{26\}}([SSq]) \approx 0.570\$$	Anchors $\$rho_{\{UA\}}\$$ in $\$Buoy_{\text{grad}}\$$; $\$Freq_{\text{open}}/r^{\{26\}}\$$ void displacement
Dipole Vortex Primes	$\$p > 26\$, \$p_{\text{special}} = 113\$$	$\$Spectra_{\text{quant}}\$$ prime vortex modes in $\$DPM_{\text{drive}}\$$; non-repeating quantum egg fingerprints
Buoyancy Harmonics	$\$U_{\{g2\}} = \sum H_m(1-e^{\{-[SSq]\}} \cdot m) \cdot \cos(\omega t_n)\$$	$\$Resonance_{\text{harm}} \rightarrow Buoy_{\text{grad}}\$$ harmonic correspondence; same damping in $\$US_{\text{egg}}\$$ integral

§6 — Validation Summary

Paper	Test	Result
PAPER_521	$\$US_{\text{range}} > 0\$$	■ $\$7.57 \times 10^{\{21\}}\$$ Hz
PAPER_522	$\$DPM_{\text{drive}} > ReRing_{\{BB\}}\$$	■ $\$\sim 10^{\{14\}}\$$ Hz
PAPER_523	Trapezoidal integral converges	■ $\$1.008 \times 10^{\{23\}}\$$ Hz·s
PAPER_524	$\$f_{\text{emerge}} \approx 0.1832\$$	■ 18.32%
PAPER_525	Hub status COMPLETE	■

§7 — Testable Predictions

1. Radio surveys should detect a $\$ReRing_{\{BB\}}\$$ background echo at $\$f \sim Freq_{\text{max}} \cdot e^{\{-1\}}\$$ in the stable-mass regime.
2. Proplyd emergence fractions in non-Orion HII regions should converge to $\approx 18\%$ if the $\$[SSq]\$$ calibration is universal.
3. The UQFF spectral tensor eigenvalue $\$\lambda_{\text{stable}}\$$ should be measurable via proplyd polarimetry in the 1/3 stable spectral band.

Cross-reference: PAPER_429 (three UQFF number systems); PAPER_521 (US spectral divisions); PAPER_522 (DPM drive); PAPER_523 (quantum egg); PAPER_524 (plasma orb)